

The patient enters the information into the microprocessor using input devices, such as keypad, touchscreen, etc. Collected data is fed to the microprocessor. Remote server is the link between the wearable device and the outer world. These communicate with each other through local area network, Internet or telemedicine facilities. Output devices provide display/feedback/alert or implement required actions, such as functional stimulation and drug delivery. Components used are miniature, light, durable, comfortable, and provide the necessary functional reliability and safety at low cost.

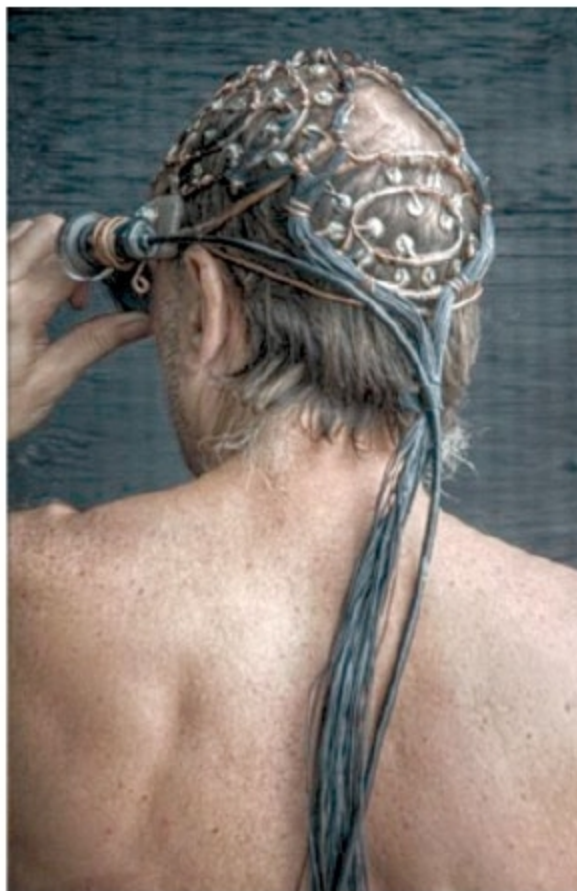
## Technologies involved

Technologies involved are discussed next.

**Computer and communication technologies.** Computer and communication technologies provide data acquisition, storage and processing, and also communication with the external world. Advanced devices use user-programmable integrated circuits, field programmable gate arrays and complex logic devices. Integration of hardware and software ensures the required processing speed.

Data storage capacity in the range of 8MB to 3GB is normally used to retain data indefinitely without battery. Input/output technologies facilitate human-computer interaction without hindering normal body activities. Technologies enable the use of such input-output devices as miniaturised microphones and speakers, voice-recognition software, body-mounted keyboards, miniaturised pointing devices, touchscreens, handheld keyboards, miniature liquid crystal or LED displays, clip-on monitors and embedded monitors in glass frames.

**Medical sensors.** Sensors provide digital electrical output that can be easily processed. These have to be wearable, non-invasive, capable of offering painless and comfortable measurement, and prevent infections and contamination. Attached to the human body, these monitor physiological or kinesiological conditions that include heart and muscle activities, respiration, skin temperature, blood pressure and body posture, movement and acceleration.



MindMesh, an EEG-based thinking cap  
(Credit: [www.interaction-design.org](http://www.interaction-design.org))

Selection of sensors is based on specific functional requirements and operating conditions. There is frequently a trade-off between size, cost and power consumption, and durability and reliability of devices in the intended operating environment.

Peripheral sensors are used to monitor information from the outer environment and include miniaturised cameras, microphones, contact sensors, environmental sensors, and location- and geographic-information-monitoring sensors.

Non-invasive sensors like electrodes are used for detecting skin surface potential in bioelectric monitoring such as ECG, electroencephalography (EEG) and electromyography (EMG). Thermistor sensors are used for monitoring skin surface temperature; piezoelectric sensors are used for monitoring heart rate and respiratory effort.

Photoplethysmography (PPG) sensors are used for detecting blood volume changes of skin area. Infrared light emitted into skin is absorbed proportional to blood volume and back-scattered corresponding to the variation of blood volume. Scattered light is measured to compute blood volume variation. Galvanic skin sensors are used for detecting skin conductance.

Oximetry sensors are used for detecting blood oxygen saturation

and heart rate. An infrared emitter/receiver worn on the finger or ear lobe transmits light through skin. Continuously varying concentrations of oxygen in blood are measured by the receiver, and heart rate is determined from the variation in concentration over time.

Several other sensors are available such as micro-electromechanical sensors (MEMS) and microelectronic biosensors (MEBS). MEMS are piezoresistive silicon pressure sensors based on Wheatstone Bridge, and are used to measure blood pressure, respiration and acceleration.

MEBS are of two types: calorimetric or electrochemical biosensors. Calorimetric biosensors are used for monitoring enzymatic reactions by detecting heat of the biological reactions. Electrochemical biosensors are used to monitor enzymes such as glucose, lactate and urea.

Potentiometric sensors are used to measure electrical potential between two electrodes at zero current flow. Amperometric biosensors are used to measure current at an electrode. Research efforts are on to produce multi-biosensor elements, integrating amperometric and potentiometric sensors on one substrate to form an integrated laboratory-on-a-chip and to develop wireless biomedical sensors.

**e-clothing and accessories.** e-clothing uses combined textile and electronic technologies. In e-clothing, electronic sensors, circuits or piezoelectric materials are attached, included or integrated into conventional clothing or accessories to store, process, retrieve and send information. Examples are conductive fabrics woven with silk fibres that have fine gold, silver or copper wrapping. A fabric keyboard that can be crushed or washed, and smart garments with integrated processors and transmitters are other examples.

Fabrics have also been developed using a micro-supercapacitor with conductive threads to store charge in self-powered smart garments. Special sewing technology creates a flexible mesh of aligned electrodes on a textile backing for a solid-state wearable device with ability to store charge that can monitor health in real time.

Examples of accessories are smart